

SangHyun Bang 방상현

Java · Spring Boot Backend Developer

[github.com/sanghyunbang](#) [betterworldwithlucas.com](#)

I find where data and state drift apart, block it with structure, and pin it down with tests. I read real SQL, query plans and logs to fix bottlenecks and structural flaws, and I check what a test actually verifies rather than trusting a green run. My business/economics background and data-analysis experience help me consider business and data flow together.

EXPERIENCE	레이시스 Backend Developer - ERP accounting module · Node.js, MariaDB Chung-Ang Univ. Industry-Academic Cooperation Foundation Student Researcher - Assisted Multi-Armed Bandit / reinforcement-learning research — theory analysis, experiment organization, project support - Built the habit of viewing problems quantitatively through experiment design and interpretation Chung-Ang University Research Intern - Bandit Algorithm / Reinforcement Learning research intern Daol Investment & Securities Management Planning Intern - Supported management-planning work — experienced how business and data connect to decisions	2026.07 - present 2023.09 - 2024.08 2023.01 - 2023.08 2022.01 - 2022.02
------------	--	---

PROJECTS	Pawloop — dog-walking & family-care app (solo) 2026.06 - present Case study ↗ ONE store ↗ Source is private · demo on request - Planned and built it alone — Flutter app, Spring Boot server and AWS deployment — and released it on ONE store (2026.09) - Per-room chat sequence — reading the room and writing +1 lets two requests read the same value and hit the per-room unique constraint. Bundled the increment and "get my number" into UPDATE ... SET last_seq = LAST_INSERT_ID(last_seq + 1): the UPDATE's row lock serializes the requests, and the incremented value stays in the session, so the number comes back without re-reading the row - Pinned with a test on real MySQL (Testcontainers): 20 threads sending to the same room get sequence numbers 1–20 exactly once each (connection pool of 10, so at most 10 concurrent transactions). Resend idempotency uses a (room_id, sender_user_id, client_msg_id) unique constraint, with three cases fixed by tests: same ID duplicated / same key from another member / no key - First production deploy (2026.09.17): Caddy exited on a config parse error and ports 80/443 did not respond — the test guarding the config only checked Caddyfile strings and passed a broken config. Replaced it with a test that runs caddy validate in the production Caddy image via Testcontainers - Modular monolith on Spring Modulith — introduced boundary checks after the fact, cut violations from 620 to 107 and locked them with a ratchet test that fails if they grow; zero dependency cycles between modules - Full backend test run: 1,994 tests, 0 failures (as of 2026-09-30). About 81% of commits (445/550) were written with an AI coding agent and are marked with commit trailers - Java 21 / Spring Boot 3.5 / Spring Modulith / JPA / MySQL 8 / Testcontainers / Docker Compose · Caddy / AWS EC2 · Terraform / Flutter BARO FARM — MSA commerce backend (member/auth, Gateway, OPA authorization) 2025.11.29 - 2026.01.27 (team project) Case study ↗ - Owned the member/auth service, the Gateway and OPA authorization in an MSA on Spring Cloud Gateway-Redis-Kafka-OPA - Diagnosed intermittent 403s as a structural "JWT permission vs user-state mismatch," not a gateway error - Designed and implemented Kafka hotlist events → OPA bundle refresh to cut the permission lag to within the polling interval (10–60s); the policy rule ran 2026-01-22 to 01-25. The policy default was switched to allow the day before the deadline, so it is inactive on main - Member-withdrawal outbox (built during the project) — written inside the withdrawal transaction, published by a scheduler, FAILED after 5 failures. No consuming service yet (design only) - Seller approval, in later solo refactoring (2026.02–03): split the admin approval path into commit-then-async propagation and recorded that decision in an ADR (unmerged branch, no tests; tokens live 7 days, so worst-case recovery is also 7 days)
----------	--

HeatTrip — Emotion-based travel recommendation (Team Lead)

2025.07 - 2025.11 (competition · team Hit디Heat, 3 people) · 2026.03 - 2026.06 (solo refactoring) [Case study ↗](#) [ONE store ↗](#)

🏆 Excellence Award, 2025 Tourism Data Utilization Competition (Korea Tourism Organization, 2025.11)

- Scoped the LLM role — for place recommendation the LLM picks only the category (cat3); actual places are filtered by cat3 and ranked inside Spring by emotion-feature, popularity and distance scores. A server path skips the LLM when cat3Filter is sent (the current app does not send it). Known limitation to fix: the emotion-input contract mismatches (app sends floats, server takes ints), so most emotion values become 0
- Split place-list pagination into Offset / Cursor (createdtime + contentid composite key, Base64 cursor, size+1 for hasNext)
- (Solo refactoring) Place search — EXPLAIN ANALYZE showed the bottleneck was the count query (full scan + repeated correlated subquery, ~755ms); switched to search_text denormalization + FULLTEXT (ngram) + MATCH...AGAINST. Per the documented plan the count full scan is gone; list ~4.9ms (single measurement for '카페')
- (Solo refactoring) Added unit tests and observability code — AOP request/exception collection, fingerprint dedup → Slack alerts, correlation id, JaCoCo coverage reports · Qodana (GitHub Actions). The observability code has no tests and no record of a sent alert yet
- Java 21 / Spring Boot 3.5 / Spring Security·OAuth2·JWT / JPA / MySQL 8 / AWS S3·CloudFront / Docker Compose

Other projects

- **SCOUT** · 2026.05 - present (solo · pre-launch) — A valuation service that estimates a fair-value range with public valuation methods (DCF · Monte Carlo). Pure Kotlin engine + Spring Boot API (PostgreSQL · Testcontainers) + Flutter app. A private project being prepared for launch; the source is not public. Not investment advice.

TRAINING	Programmers Dev Course · MSA Spring AI Intensive Spring Cloud · Spring AI · MSA backend — BARO FARM project	2025.11 - 2026.01
	Gangnam HiMedia · KDT Full-stack Bootcamp Spring · React full-stack — Excellence Award for the graduation project (OREUM, hiking community)	2025.01 - 2025.07
EDUCATION	Academic Credit Bank System B.Eng. in Computer Engineering · 4.05 / 4.5 · conferred 2026.08	2025.07 - 2026.08
	Sogang University Business Administration & Economics, double major · Summa Cum Laude · 3.75 / 4.3	2015.03 - 2023.02
	Chung-Ang University Graduate School M.S. program in Artificial Intelligence · on leave	2023.09 -
SKILLS	Backend — Java, Kotlin, Spring Boot, Spring Security, Spring Cloud Gateway, Spring Modulith, JPA / Hibernate, Node.js DB / Messaging — MySQL, MariaDB, PostgreSQL, Redis, Kafka Infra / Test — Docker Compose, Caddy, AWS (EC2 · S3 · CloudFront), Terraform, OPA, Testcontainers, GitHub Actions Data / AI — Python, LLM integration, Data Analysis, Reinforcement Learning Mobile / Etc. — Flutter, Git	
CERTIFICATIONS	• Engineer Information Processing (2026.06) • SQL Developer (SQLD) (2026.06) • ADsP — Data Analysis Semi-Professional (2022.09) • Certified Investment Manager (2022.12) Language — IELTS 7.0 (2024.12) · OPIc IH (2026.03)	
AWARDS	• Excellence Award, 2025 Tourism Data Utilization Competition (Korea Tourism Organization, 2025.11) — HeatTrip • Excellence Award, KDT Full-stack Bootcamp graduation project (Gangnam HiMedia, 2025.07) — OREUM • Honorable Mention, 13th DB FEC (DB Kim Jun-ki Cultural Foundation, 2023.05) • Bank of Korea Monetary Policy Competition — preliminary round, top 11 teams in Seoul (2022.07)	